

RESEARCH STATEMENT

Hanbaek Lyu

I work in discrete probability and in machine learning and AI, and I publish in the leading venues of both fields. My research comprises 49 published or accepted papers, with 8 more under review. In probability, my papers appear in top venues including the *Annals of Probability*, the *Annals of Applied Probability*, *Forum of Mathematics*, *Sigma*, *Transactions of the AMS*, and *IMRN*. In machine learning and AI, these include 17 papers in the field’s top venues—8 at *ICML*, 3 at *NeurIPS*, 5 in *JMLR*, and 1 in *SIAM Journal on Optimization*—along with a *Nature Communications* paper featured as an Editor’s Highlight. My research has been supported by **three NSF grants** (DMS-2010035, 2206296), most recently an **NSF CAREER Award** (DMS-2541878, \$450,000, 2026–2031) on random matrices, Schrödinger bridges, optimal transport, and generative modeling, and also by an industry funding from KRAFTON AI on developing linear attention for long-context language models and Schrödinger-bridge-based generative modeling.

1. Discrete Probability: EMERGENCE IN LARGE-SCALE SYSTEMS

In discrete probability, I establish scaling limits, phase transitions, and universality classes for large discrete systems. A recurring theme of my work is the discovery of unexpected connections between seemingly distant objects. Synchronization of coupled oscillators is connected to clustering in an annihilating particle system [ALS24], and its long-term behavior is governed by the persistence probabilities of a random walk with correlated increments [LS19]. Three-color excitable media can be viewed as a projection of Lamport’s classical clock synchronization algorithm. A two-type diffusion-limited annihilating system can be analyzed through embedded token-passing and path-swapping processes [JLS23]. The longest soliton lengths in randomized box-ball systems are connected to the extreme behavior of a circular exclusion process and to its diffusion limit, a semimartingale reflecting Brownian motion [LLP22, LLPS24]. And random matrices with given margins concentrate around static Schrödinger bridges.

1.1. Random Matrices with Given Margins and Entropic Optimal Transport. Large random matrices with prescribed row and column sums (margins) arise in contingency tables, network flows, and graphs with given degree sequences, and they exhibit phase transitions not seen in classical models. As Barvinok conjectured in 2010, a small change in the margins of a random contingency table can trigger a sharp phase transition—a non-analytic change in its typical structure that suggests computational hardness. With Dittmer and Pak, I settled this conjecture in *Transactions of the AMS* [DLP20] (Figure 1), and with Pak, I showed that Good’s 1950 independence heuristic for counting contingency tables fails dramatically beyond the transition [LP22].

With Mukherjee, I then found the principle underlying these phenomena. Our key innovation is a *transference principle*, which approximates a margin-conditioned matrix by a tractable matrix with independent entries drawn from an exponential tilt of the base measure. The surprise is that the required tilt is given by the two Schrödinger potentials of an implicit static Schrödinger bridge, that is, an entropic optimal transport problem. This framework brings the tools of entropic optimal transport—Kantorovich duality, Schrödinger potentials, and Sinkhorn’s algorithm—to previously intractable problems on random matrices with given margins. This work [LM25] is under review in *Probability Theory and Related Fields*.

1.2. Integrable Systems and Solitons. Box-ball systems (BBS) are discrete integrable systems that originate from the Korteweg–de Vries (KdV) equation, crystallizing its classical soliton solutions in a combinatorial setting. I study how solitons emerge and evolve in such systems from a disordered random initial state. My

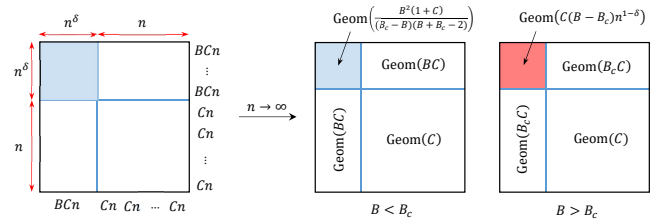


Figure 1. Phase transition in random contingency tables [DLP20]. (Left) A table whose first $\lfloor n^\delta \rfloor$ rows and columns have margins $[BCn]$ and whose remaining n rows and columns have margins $[Cn]$. (Right) Limiting distributions of the entries of the uniform table in the subcritical ($B < B_c$) and supercritical ($B > B_c$) regimes, for $1/2 < \delta < 1$; $\text{Geom}(\lambda)$ is the geometric distribution with mean λ .

work with Levine and Pike in *IMRN* [LLP22] is a foundational paper on randomized BBS, establishing the scaling limit of soliton lengths in the 1990 Takahashi–Satsuma BBS with i.i.d. initial configurations.

With Lewis, Pylyavskyy, and Sen, in *Forum of Mathematics, Sigma* [LLPS24], I established the phase diagram and precise asymptotics of the top soliton lengths in the multicolor BBS. The analysis centers on the *multicolor carrier process*, a high-dimensional Markov chain in the integer orthant that behaves like a complex queueing system. By establishing its scaling limit and ruin probabilities, we obtained asymptotics for the soliton lengths down to the $\log \log n$ term, a precision beyond the reach of the thermodynamic Bethe ansatz. More recently, with Keating, Kim, and Loeser [KKLL25], I constructed a stochastic BBS that interpolates between the deterministic BBS, a discrete counterpart of the KdV equation, and PushTASEP, a particle system in the KPZ universality class, and proved its convergence to a semimartingale reflecting Brownian motion in the diffusive limit.

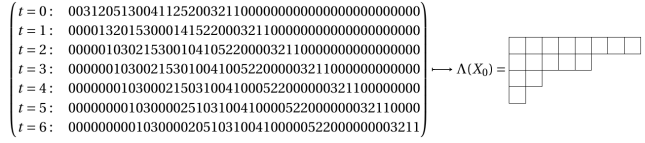


Figure 2. Evolution of a 5-color box-ball system, which decomposes into solitons of nondecreasing lengths. The soliton lengths are encoded by the invariant Young diagram on the right.

2. Optimization and Machine Learning: ALGORITHMS TO THEORY TO APPLICATIONS

My research in optimization and machine learning centers on matrix factorization and its generalization to nonconvex constrained optimization. For these problems I develop principled algorithms with rigorous convergence guarantees, including online matrix and tensor factorization for Markovian data [LNB20, LSN22], supervised matrix factorization [LLY23, LLY24], stochastic first-order methods for correlated data [Lyu24, AL23, PL24], and block majorization-minimization [LL25] with its Riemannian extensions [LBNL26, LBNL24]. My work also bridges theory and practice by explaining why widely used learning algorithms work, including cyclic block optimization for nonconvex problems [LL25], policy optimization in reinforcement learning [PKXL25, PKXL26], and coordinate-ascent maximum likelihood for phylogenetic inference [CJLRS26, CJLRS25, CLR25].

2.1. Cyclic Block Optimization. Many large-scale optimization problems are solved by cyclic block optimization: one optimizes a single block of parameters, then moves to the next block, cyclically. Despite its simplicity, broad applicability, and proven practical efficacy, its theory has lagged behind that of the random coordinate methods pioneered by Nesterov, Polyak, and Wright. With my student Li, I developed a new analysis that settled the convergence rate of such methods in the most challenging and practical setting, that of nonconvex, nonsmooth, constrained problems; this work appeared in the *SIAM Journal on Optimization* [LL25], one of the top two journals in optimization. We further extended these results to Riemannian manifolds, which capture intricate geometry in the parameter space [LBNL26, LBNL24] (in *ICML* and *JMLR*). This theory gave the first iteration-complexity bound for the Bures–JKO scheme for Wasserstein variational inference with non-strongly convex potentials.

2.2. Stochastic Optimization. For large-scale data streams, in which data arrive sequentially in real time, one needs stochastic learning algorithms that process the data sequentially and gradually learn the parameter that best explains the stream. A major challenge is that real-world data streams, such as web crawls and network samples, violate the i.i.d. assumptions on which optimization theory relies. In a series of *JMLR* papers [LNB20, LSN22, Lyu24], I developed a general family of stochastic block majorization-minimization algorithms and proved convergence guarantees for non-i.i.d. data. This framework unifies a wide range of classical and modern algorithms, including Nesterov’s dual averaging, proximal-point empirical loss minimization, online expectation-maximization, and online matrix and tensor factorization.

2.3. Machine Learning Applications. Building on my work in discrete probability and nonconvex optimization, I also develop machine learning methods with broad impact. I have developed methods for decomposing networks into a small set of key connection patterns and reconstructing networks from them (Figure 3). The culmination of this work appeared in *Nature Communications* as an Editor’s Highlight [LKVP24], where a referee called it a “tour de force.” The core idea is to sample many subgraph patterns using random k -paths [LMS23] and to extract a small number of essential patterns by online matrix factorization [LNB20]; the extracted patterns reveal how a network is composed at the mesoscale. One application is learning single-molecule chromatin interaction patterns [PPL⁺]. Using supervised matrix factorization, I have also developed methods for identifying latent gene groups associated with specific cancers [LLY23, LLY24] (in *NeurIPS* and *ICML*), which recover clinically known oncogenes.

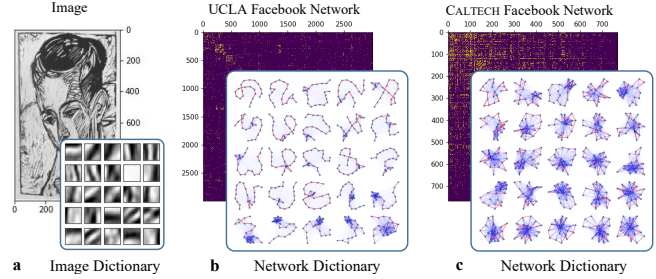


Figure 3. Dictionaries learned from square patches of an image (a) and from 20-node subgraphs of the UCLA (b) and CALTECH (c) Facebook networks [LKVP24].

3. CURRENT AND FUTURE RESEARCH PROGRAM

My research over the next five years is organized around three directions. The first is the program of my NSF CAREER Award on random matrices, Schrödinger bridges, optimal transport, and generative modeling; the second carries my work on solitons and box-ball systems toward the central questions of integrable probability; the third is an industry collaboration with KRAFTON AI on the architecture of large language models.

3.1. Random Matrices, Schrödinger Bridges, Entropic Optimal Transport, and Generative Modeling. Many large systems are observed only through two marginal views. A traffic planner may know how many cars leave each neighborhood and how many arrive in each district, but not which routes connect them; a model asked to turn sketches into photographs may see many sketches and many photographs, but no examples pairing the two. In each case the totals along the edges of a table are known, while the entries that tie the two views together are hidden. One can fill in the hidden table in two natural ways: at random, in the most unbiased way consistent with the margins, or by design, so that the transport between the two views is as efficient as possible. The central message of my CAREER program (DMS-2541878, 2026–2031) is that these two answers—one shaped by chance, the other by optimality—are remarkably close: margin-constrained random matrices are governed by a hidden variational principle, the static Schrödinger bridge of entropic optimal transport. Building on this insight, the program develops a unified theory of random and optimal transport and turns it into new methods for generative modeling, along three aims: (1) the foundational theory of generalized Schrödinger bridges, including their Kantorovich duality and the convergence of Sinkhorn-type algorithms; (2) probabilistic techniques that transfer results

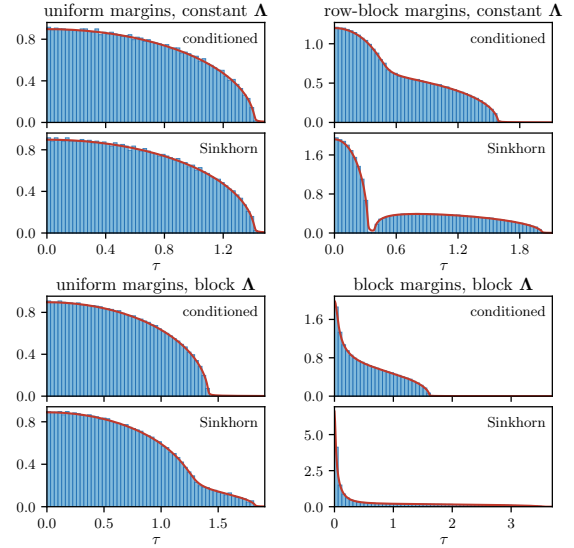


Figure 4. Singular values of the fluctuation matrix (histograms) and the density predicted by the matrix Dyson equation (red) for 5000×5000 tables, under four choices of margins and prior mean. In each pair, the upper panel shows a Poisson matrix conditioned on its margins [DLP26a], and the lower panel shows the same Poisson matrix rescaled to the same margins by Sinkhorn’s algorithm [DLP26b]. The two models share a typical table but have different fluctuation spectra, both governed by the matrix Dyson equation.

from tractable, exponentially tilted models to otherwise intractable margin-constrained random matrix ensembles; and (3) new generative methods, including multi-modal image generation via Schrödinger bridges with domain-adaptive divergences and random transport.

Two recent papers with my students Duan and Powell develop the random-matrix side of this program. For Sinkhorn-rescaled random matrices, we established a quantitative stability theory for the static Schrödinger bridge and used it to prove scaling limits and fluctuation results, including bulk rigidity of the spectrum and a central limit theorem for the empirical Schrödinger potentials [DLP26b]. For Poisson random matrices conditioned on their margins, we proved a transference principle to independent entries, a law of large numbers toward the Schrödinger bridge, and a central limit theorem; through a representation by Mallows-type random permutations, these results yield Pal’s conjecture on their limiting partition function under weaker assumptions, as well as a conjecture of Mukherjee [DLP26a]. The two models share a typical table but have different fluctuation spectra, both governed by the matrix Dyson equation (Figure 4). On the algorithmic side, with my student Bi and with Jakwang Kim, I am developing stochastic Sinkhorn-type algorithms for entropic optimal transport with Lagrangian costs, such as shortest-path distances on large graphs, for which the Gibbs kernel cannot even be formed: a stochastic mesoscale approximation replaces the kernel by an average of cheap local kernels on randomly sampled blocks, and its accuracy reduces to the kernel stability of the Schrödinger bridge [BKL26].

The third aim connects this theory to generative modeling. A Schrödinger bridge between two data distributions translates one into the other without paired examples—the sketch-to-photograph problem above. In my 2025 summer REU program on generalized Schrödinger bridges, undergraduate students implemented diffusion Schrödinger bridges for unpaired image translation (Figure 5). The next step is to replace the KL divergence by divergences adapted to each data domain—for which Sinkhorn-type algorithms still converge linearly [CL25]—and to extend the framework to multi-marginal settings such as video generation. This direction is also supported by industry funding from KRAFTON AI.



Figure 5. Unpaired Dog→Human translation using a dynamic Schrödinger bridge. Image produced in my 2025 summer REU program on diffusion modeling, supported by NSF DMS-2206296.

3.2. Box-Ball Systems and Integrable Probability. Outside the CAREER program, I am pushing my work on box-ball systems toward the central questions of integrable probability. For the deterministic box-ball system with random initial data, the fluctuation theory of solitons is now well developed: Olla, Sasada, and Suda showed that a tagged soliton converges to Brownian motion in the diffusive scale and satisfies a large deviation principle [OSS24]. My program asks what happens when a small amount of stochasticity is introduced into the dynamics itself (Figure 6). I am developing the fluctuation theory of such stochastic box-ball systems from two perspectives: invariance principles, and the algebraic methods of KPZ fluctuation theory. With Keating, Kim, and Loeser [KKLL25], I established a general invariance principle for d -dimensional Markov jump processes converging to semimartingale reflecting Brownian motions in convex polyhedra. It accommodates distinct boundary dynamics on every face intersection, and hence an overdetermined Skorokhod decomposition with more reflection vectors than faces. Applied to the stochastic box-ball system, a probabilistic interpolation between the Takahashi–Satsuma system and PushTASEP, it yields a diffusive limit whose reflection matrix records a “solitonic bias” absent from PushTASEP, together with a unified diffusion limit for exclusion processes with asymmetric collisions.

With Aguirre, Kim, and Paguyo [AKLP26], I examined the second perspective. Box-ball systems with a stochastic carrier, which fails to pick up or drop off balls with prescribed probabilities, form a family of stochastic higher-spin vertex models, and the coordinate Bethe ansatz is the entry point to the algebraic

methods behind known KPZ fluctuation scaling results. Here the answer is negative. We classified the stochastic weights for which the coordinate Bethe ansatz closes at every particle number: they are exactly the fusion weights of Corwin and Petrov, a subfamily that stays a uniformly positive distance from the box-ball point at every capacity. In particular, the ansatz does not close for the stochastic box-ball system at any capacity $J \geq 2$, the regime in which solitons exist: plane-wave solvability fails strictly before the crystal limit, where Yang–Baxter integrability survives only in combinatorial form.

These results set the agenda for the next five years. The invariance principle applies to a wide range of particle systems but does not use the integrable structure of the box-ball system, while the coordinate Bethe ansatz uses that structure but does not survive a stochastic carrier. My aim is a fluctuation theory between the two, along three lines. First, I will extend the invariance principle to a number of particles that grows with the diffusive scaling, where the limit is an infinite-dimensional reflected diffusion, and determine whether these limits carry a random-matrix structure, as Warren’s reflected interlacing Brownian motions do for the GUE corners process and, by the work of Gorin and Shkolnikov, for multilevel TASEP and the edge of multilevel Dyson Brownian motion [War07, GS15, GS14]. Second, the algebraic route remains open beyond plane-wave closure: our analysis produced exact formulas for these vertex models, including exact recursions for the generating function at capacities $J = 2$ and 3 and a closed form for the top Bethe amplitude at every capacity, and I intend to build on them an alternative asymptotic analysis adapted to the near-box-ball corner of the lattice models. Third, the stochasticity injected in current models may be too severe to preserve the combinatorial integrability of the box-ball system; I aim to identify stochastic perturbations mild enough to retain it, so that their fluctuations can be analyzed with the combinatorial tools of the deterministic system.

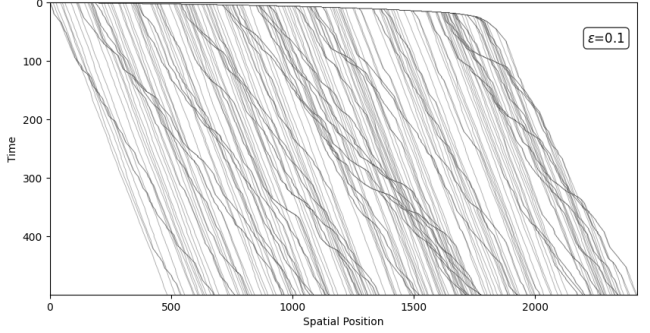


Figure 6. Ball trajectories in the stochastic box-ball system with infinite capacity and $\varepsilon = 0.1$, started from a block of 200 consecutive balls. The initial soliton breaks apart because the carrier occasionally fails to pick up balls; the long-term dynamics are governed by singleton balls performing independent random walks and by the spontaneous emergence of short solitons.

3.3. Machine Learning with KRAFTON AI. Through an industry collaboration with KRAFTON AI, I work on the architecture of large language models, where tools from the first direction find new applications. The first project addresses long-context language modeling with linear attention, which replaces softmax attention by a recurrence on a fixed-size state and thereby trades the growing key–value cache for memory collisions. We introduced Integral DeltaNet (Figure 7), which pairs a fast delta-rule state holding current key–value associations with a slow integral state retaining a weighted history of it, combines the two adaptively at readout, and admits a chunkwise-parallel formulation for hardware-efficient training. At 340M and 1.3B parameters, trained on 15B and 100B tokens, it improves on the Kimi Delta Attention and Gated DeltaNet-2 baselines in language modeling, commonsense reasoning, and retrieval.

The second project, with my student Moon, studies Sinkhorn attention, in which softmax normalization is replaced by a doubly stochastic Sinkhorn rescaling, so that each attention matrix is a static Schrödinger bridge between uniform marginals. For i.i.d. tokens, we show that the Sinkhorn attention computed from finitely many tokens is close, with high probability, to the population-level scaling evaluated at the sampled

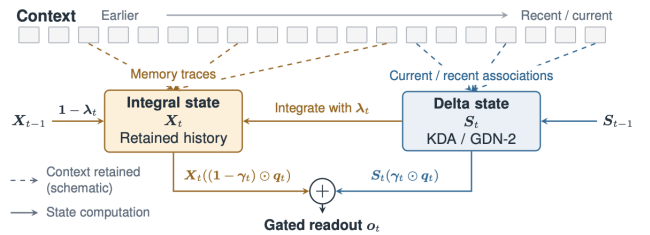


Figure 7. Integral DeltaNet. A fast delta state holds current key–value associations, and an integral state accumulates a weighted history of it; at readout, the model adaptively combines the two to access information across timescales.

tokens, using the potential-stability theory of [DLP26b], and we are removing the boundedness assumption on the token embeddings. The longer-term aim is a random-matrix theory of attention: fluctuation and spectral theory for Sinkhorn-rescaled kernel matrices, and a principled account of what fixed-size linear-attention states can and cannot retain.

These directions admit entry at several levels: generative-modeling experiments suit undergraduate researchers, as in my REU programs; the algorithmic and box-ball questions are concrete enough for a second-year graduate student; and the transference and fluctuation theory call for a postdoc or an established collaborator. I have supervised five doctoral students, two postdoctoral scholars, and twenty-four undergraduate researchers across five REU cohorts, and I structure groups so that postdocs mentor graduate students and graduate students mentor undergraduates.

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